

Reporting— Agriculture

Olive trees, hope for Pakistani farmers in the face of climate change



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Olive cultivation is attracting more and more farmers in Pakistan. Resilient and adapted to arid lands, it is becoming essential in this country, which is among the most exposed to climate change.

Nowshera District (Pakistan), report

" *Before 2014, it was just wasteland* ," says Salahuddin, in his fifties, pointing to the rows of olive trees. They stretch as far as the eye can see from his farm in northern Pakistan. Dressed in a crisp white kurta and flanked by his four children, the farmer reaches into a drum and pulls out a handful of macerating olives.

For the past ten years, Salahuddin has been a pioneer. On his 17-hectare plot in the Nowshera district of northern Pakistan, he cultivates nearly 8,000 olive trees for oil production. This crop has been relatively uncommon in Pakistan, where the native species, in the wild, produces only very small fruits unsuitable for oil production. " *This region is ideal for olive trees; there is a huge amount of suitable virgin land* ," the farmer enthuses, raising a fruit to his lips, hidden by his salt-and-pepper beard.



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Pakistanis are welcoming this new crop even more since their country was the hardest hit by climate change in 2022, according to the NGO Germanwatch . Olive trees, which are particularly resilient in arid and rugged terrain and do not require as much water as other

plants, are therefore a source of hope for the country's farmers.

Some even go so far as to speak of a real " *fashion phenomenon* ", driven by constantly increasing demand and encouraged by good results: there were more than 15,000 producers in the country in 2024, compared to only 7,000 in 2021.

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To maximize the tree's potential in the face of climate change, Salahuddin, a proponent of natural methods, practices intercropping with citrus trees. " *These two plants complement each other very well* ," he explains. " *The olive tree protects the citrus trees from the sun, and in turn, the citrus trees allow the olive tree to better withstand the heat thanks to their dense canopy, which creates a cooler and more humid microclimate. Even when it's 44°C, the temperature in the orchard doesn't exceed 39°C.* " Thanks to subsidies from the Pakistani government, the farmer installed a drip irrigation system, which helps avoid water waste.



Salahuddin, who cultivates nearly 8,000 olive trees in northern Pakistan. © Ondine de Gaulle / Reporterre

Salahuddin's olive seedlings were provided to him free of charge as part of the Italian-Pakistani cooperation initiative (Pidsa). It was this plan that kick-started the development of an olive oil sector in Pakistan in 2012. The farmer also received training in Italy at the International Centre for Advanced Mediterranean Agronomic Studies (CIHEAM) in Bari.

6 million trees planted

Through these support measures, Pakistan seeks to reduce its heavy dependence on imported cooking oils, stimulate the local economy, and ultimately become an olive oil exporter. Since the project's launch, nearly 6 million trees have been planted across all provinces of the country, covering some 25,000 hectares of land, much of it previously undeveloped.

This represents a considerable opportunity for many farmers struggling to make ends meet. Pakistan's agriculture, which is not very diversified, relies heavily on low-yielding staple crops compared to other countries, such as wheat and rice, which are particularly vulnerable to water stress and climatic hazards. This year, Salahuddin made a profit of 20 million Pakistani rupees (€60,000).



The number of olive growers in Pakistan has more than doubled in three years. © Ondine de Gaulle /

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" This season, we produced 7,500 liters of olive oil, compared to 5,000 last year. And this orchard could produce up to 10,000 !" he enthuses. *"It's a cash crop: once the oil is extracted, demand is very high. "* He currently sells his oil to the public consortium Pak Olive, a brand created in 2023, while preparing to launch his own brand. On the local market, however, olives remain a luxury product. Without a significant increase in production in the coming years, they will remain out of reach for most households.

Farmers can extract their oil at government-run mills provided free of charge. In Nowshera, the mill, inaugurated in 2015, is equipped with machines capable of pressing up to 500 kg of olives per hour. *" Now that farmers have been trained in olive cultivation, the main challenge is to open a laboratory to study plant diseases, insect pests, and soil nutrition to improve production , "* explains Muhammad Nawaz, a research entomologist at the Cereal Crops Research Institute, which oversees the mill.



Muhammad Nawaz, in the government mill of Nowshera. © Ondine de Gaulle / Reporterre

Another advantage: the carbon footprint of this crop is very low. *" One hectare of olive trees can sequester 5 tons of CO₂ per year. This is less than species such as poplar,*

eucalyptus and leucaena, which can capture between 8 and 15 tons. But olive trees can live for several hundred years and thus ensure carbon storage over much longer periods , ” says Mohammad Arabi Awan, a scientific expert with the Olive Culture project.

This program, led by Ciheam of Bari in partnership with the Pakistani Ministry of National Food Security, is conducting an in-depth study on the carbon footprint of olive orchards.

“ It’s a true treasure of nature .”

The study will serve as a roadmap for Pakistani authorities to define a framework that promotes olive cultivation as a lever for mitigating climate change. *” It could also pave the way for the sale of carbon credits on the*

international market , ” explains the agricultural engineer. This is a potential additional source of income for farmers, all the more relevant in a country like Pakistan, where emissions from the agricultural sector remain low due to underdeveloped mechanization.

For Mohammad Arabi Awan, the olive tree is much more than just a tree: *” It is a true treasure of nature, at once a pillar of food security, a reservoir of nutraceutical and medicinal benefits, and a resilient crop offering sustainable livelihoods. ”* Israr, a farmer from the Lower Dir district, even dedicated a poem in Pashto, the local language, to this ancient tree, cited several times in the Quran, which he holds up as a sacred and resilient symbol:

It grows among the gardens of the Almighty and flourishes on the earth blessed by His presence. Allah has exalted it in His sacred verses and makes life spring forth from abandoned soils.

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